WORKED SOLUTION

STEP I 2018, Question 13

Probability and Statistics • Binomial and Bayes

STEP I 2018, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q13

The key idea

Everything rests on one count: if a candidate attempts k questions and gets j of them right, the score is $2j-(k-j)=3j-k$. So passing means getting at least $(k+5)/3$ right, and for each k this tells us exactly which binomial terms to add up. In part (iii) the neat move is to treat each question as having three outcomes (right, wrong, not attempted), which turns the problem into a single multinomial count.

Solution

Throughout, let $p=\frac{1}{n}$ be the probability that a randomly chosen answer is correct, and $q=1-p$. Since $n\geq 2$, we have $0<p\leq\frac{1}{2}$.

Part (i)

Let P_k be the probability that A passes when she attempts exactly k questions. Her choices are independent, so the number J she gets right has the binomial distribution $\mathrm{B}(k,p)$. Her score is $3J-k$, so she passes exactly when

$$3J-k\geq 5 \iff J\geq\frac{k+5}{3}.$$

Going through the values of k:

- $k=0,1,2$: we would need $J\geq\frac{5}{3},2,\frac{7}{3}$ respectively, so $J\geq 2$, $J\geq 2$ and $J\geq 3$, which are impossible with $J\leq k$ (the maximum score is $2k\leq 4$). So $P_0=P_1=P_2=0$.
- $k=3$: we need $J\geq\frac{8}{3}$, so $J=3$, and $P_3=p^3$.
- $k=4$: we need $J\geq 3$, so

$$P_4=\binom{4}{3}p^3q+p^4=4p^3(1-p)+p^4=p^3(4-3p).$$

- $k = 5$: we need $J \geq \frac{10}{3}$, so $J = 4$ or 5, and

$$P_5 = \binom{5}{4} p^4 q + p^5 = 5p^4(1-p) + p^5 = p^4(5-4p).$$

Now compare P_4 with its two rivals. Since $p < 1$, we have $4 - 3p > 1$, so $P_4 > p^3 = P_3$. Also

$$P_4 - P_5 = p^3\big[(4-3p) - p(5-4p)\big] = p^3\big(4p^2 - 8p + 4\big) = 4p^3(1-p)^2 > 0,$$

because $0 < p < 1$. As $P_4 > 0$ beats $P_0 = P_1 = P_2 = 0$ too, P_4 is strictly the largest, so A should choose $k = 4$, as required.

Ask yourself Isn't attempting all five questions bound to give more chances to score?

Not here. With five attempts she needs four right, because a fifth attempt costs a mark whenever it is wrong, and it is wrong far more often than right. With four attempts, three right is enough. The factorisation $4p^3(1-p)^2$ shows the advantage for every $n \geq 2$, not just for large n.

Part (ii)

Let K be the number of questions B attempts, so $\mathrm{P}(K = k) = \frac{1}{6}$ for $k = 0, \ldots, 5$, and let S be the event that B passes. Given $K = k$, B is in exactly A's position, so $\mathrm{P}(S \mid K = k) = P_k$. By Bayes' theorem,

$$\mathrm{P}(K = 4 \mid S) = \frac{\frac{1}{6}P_4}{\frac{1}{6}(P_0 + P_1 + \cdots + P_5)} = \frac{p^3(4-3p)}{p^3 + p^3(4-3p) + p^4(5-4p)}.$$

Cancelling p^3 (which is non-zero), the denominator becomes $1 + 4 - 3p + 5p - 4p^2 = 5 + 2p - 4p^2$, so

$$\mathrm{P}(K = 4 \mid S) = \frac{4-3p}{5+2p-4p^2}.$$

Putting $p = \frac{1}{n}$ and multiplying top and bottom by n^2:

$$\mathrm{P}(K = 4 \mid S) = \frac{4n^2 - 3n}{5n^2 + 2n - 4} = \frac{n(4n-3)}{5n^2 + 2n - 4}.$$

Part (iii)

For each question, C's coin toss and her choice of answer are independent of each other and of what happens on the other questions. So each question independently has one

of three outcomes:

$$\begin{aligned} \text{correct } (+2): &\quad \frac{n}{n+1}\cdot\frac{1}{n} = \frac{1}{n+1}, \\ \text{wrong } (-1): &\quad \frac{n}{n+1}\cdot\frac{n-1}{n} = \frac{n-1}{n+1}, \\ \text{not attempted } (0): &\quad \frac{1}{n+1}. \end{aligned}$$

Suppose she gets c right and w wrong, with $c+w \le 5$. She passes when $2c - w \ge 5$. This forces $c \ge 3$, and then:

- $c = 3$: we need $w \le 1$;
- $c = 4$: $w \le 1$ automatically, since $c + w \le 5$;
- $c = 5$: $w = 0$.

The number of ways of assigning c correct, w wrong and $5 - c - w$ unattempted questions is the multinomial coefficient $\frac{5!}{c!\,w!\,(5-c-w)!}$. Since a correct and an unattempted question have the same probability $\frac{1}{n+1}$, each arrangement has probability $\frac{(n-1)^w}{(n+1)^5}$. Adding up the five cases:

(c, w)	arrangements	probability $\times (n+1)^5$
$(3,0)$	10	10
$(3,1)$	20	$20(n-1)$
$(4,0)$	5	5
$(4,1)$	5	$5(n-1)$
$(5,0)$	1	1

The total is $16 + 25(n-1) = 25n - 9$, so

$$\mathrm{P}(\text{C passes}) = \frac{25n-9}{(n+1)^5}.$$

Checks

- With $n = 2$: $P_3 = \frac{1}{8}$, $P_4 = \frac{5}{16}$, $P_5 = \frac{3}{16}$, so $k = 4$ wins. In (ii) the formula gives $\frac{2\cdot 5}{20} = \frac{1}{2}$, matching $\frac{5/16}{10/16}$ directly.
- For (iii) with $n = 3$ the answer is $\frac{66}{1024} \approx 0.0645$, and a computer simulation of 200 000 tests gave 0.0644. All three formulas also agree with exact enumeration by computer for $n = 2, \ldots, 29$.
- As $n \to \infty$ the answer to (ii) tends to $\frac{4}{5}$: when guessing is hopeless, a pass almost always comes from four attempts (needing three right) rather than five (needing four right) or three (needing all three right, which has probability p^3, much less than $4p^3$).

Common slips

- Forgetting that with five attempts three right gives $6 - 2 = 4$, which fails: you need four right.
- In (ii), leaving P_3 out of the denominator: attempting three and getting all three right is also a way to pass.
- In (iii), treating it as "attempt k questions with probability $\binom{5}{k}h^k(1-h)^{5-k}$" and then using part (i) works, but it is easy to lose a term; the three-outcome count is cleaner.

Beyond STEP

Part (iii) is an example of *thinning*: each question is kept with some probability and the outcomes stay independent, so the counts of right, wrong and skipped answers have a multinomial distribution. Part (ii) is Bayesian updating, the idea behind much of modern statistics. Both are developed in Blitzstein and Hwang, *Introduction to Probability*, and in Grimmett and Welsh, *Probability: An Introduction*.

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